

EXPERIMENT 7

Develop a program to read the pH value of a various substances like milk, lime and water.

Components Required

- Drag **Arduino UNO** onto the workspace.
- Drag a **Potentiometer**
- Drag a **Breadboard** (optional, for neatness)
- Add **jumper wires**

Potentiometer Pin	Connect To
Left pin	GND on Arduino
Middle pin	A0 (Analog pin A0)
Right pin	5V on Arduino

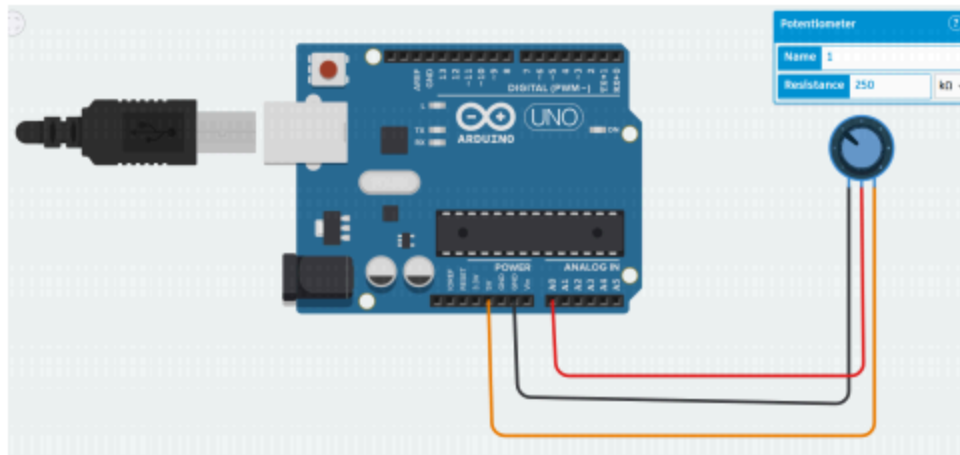
STEPS

- a. Open Arduino IDE then goto tools and select appropriate Arduino board.
- b. Select necessary components and make circuit connections as shown in the diagram
- c. Make sure VCC and Ground pins connected properly to avoid any damage to Arduino board.
- d. write the code and run a simulator to get appropriate results.



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Circuit diagram



Code

```
void setup()
{
  Serial.begin(9600); // Initialize Serial Monitor
}

void loop() {
  int sensorValue = analogRead(A0); // Read analog input
  float voltage = sensorValue * (5.0 / 1023.0); // Convert to voltage
  float pH = 7 - ((2.5 - voltage) / 0.18); // Calculate pH

  Serial.print("Voltage: ");
  Serial.print(voltage);
```



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```

Serial.print(" V -> pH: ");
Serial.println(pH);

// Check which substance it likely is
if (pH >= 1.5 && pH <= 2.5) {
    Serial.println("Substance Detected: Lime Juice (Highly Acidic)");
}
else if (pH > 6.0 && pH <= 6.8) {
    Serial.println("Substance Detected: Milk (Slightly Acidic)");
}
else if (pH > 6.8 && pH <= 7.2) {
    Serial.println("Substance Detected: Water (Neutral)");
}
else {
    Serial.println("Unknown Substance");
}

Serial.println("-----");
delay(2000); // Wait 2 seconds before next reading
}

```

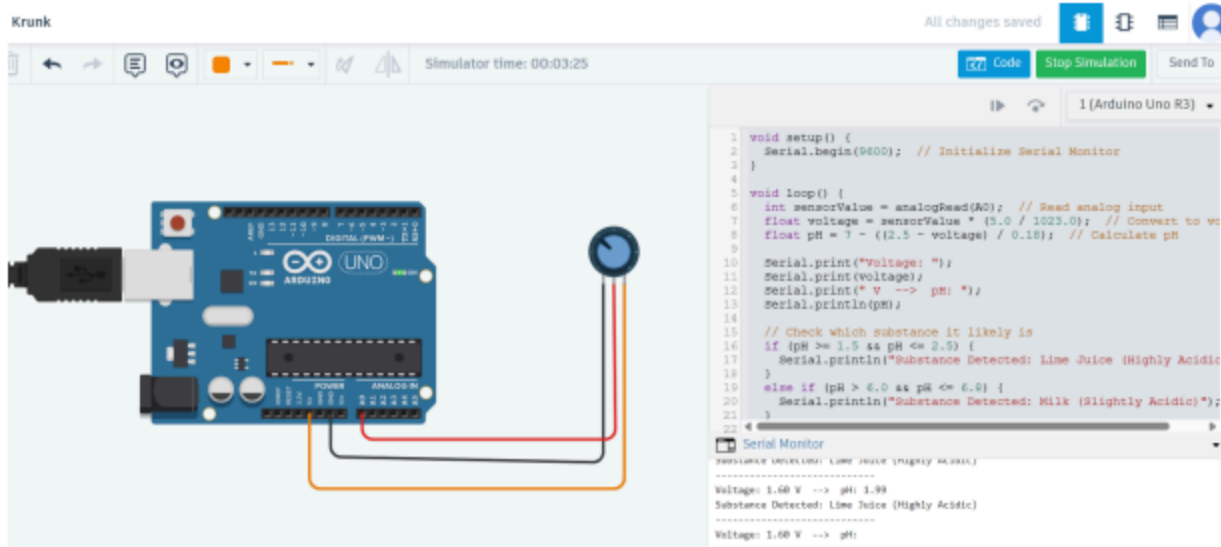
Output

- Open the **Serial Monitor**
- Rotate the potentiometer to simulate different pH values:
 - Near **2.5 V** → pH ≈ 7 (water)
 - Near **2.3 V** → pH ≈ 6.5 (milk)
 - Near **1.6 V** → pH ≈ 2 (lime)

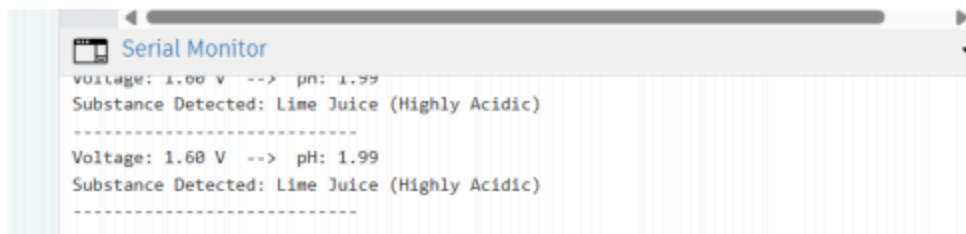


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- A real pH sensor (like the PH-4502C) gives an **analog voltage output**.
- Voltage ranges:
 - pH 7 (neutral) → ~2.5V
 - pH < 7 (acidic) → voltage drops (e.g., pH 2 → ~1.6V)
 - pH > 7 (basic) → voltage increases (e.g., pH 10 → ~3.2V)



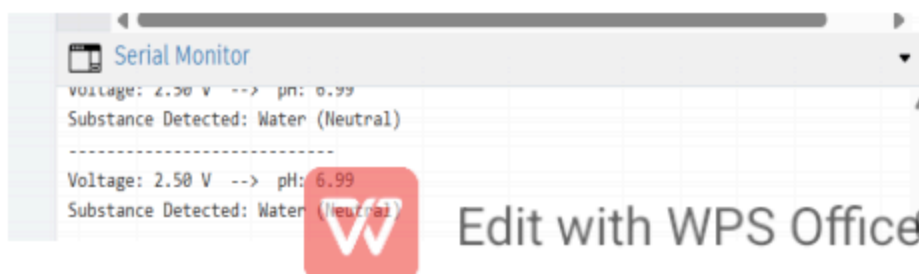
For Lime Juice



For Milk



For Water



Experiment 8

Develop a program to detect the gas leakage in the surrounding environment.

Components Required

1. Arduino Uno
2. MQ-2 Gas Sensor
3. LED (optional – to indicate leakage)
4. Resistor (220Ω for LED)
5. Breadboard
6. Wires

To detect gas leakage in the surrounding environment using Tinkercad, you can simulate it using an MQ-2 gas sensor and an Arduino Uno. The MQ-2 sensor detects gases like LPG, smoke, methane, and propane. Tinkercad allows simulating this behavior with a simple circuit and code.

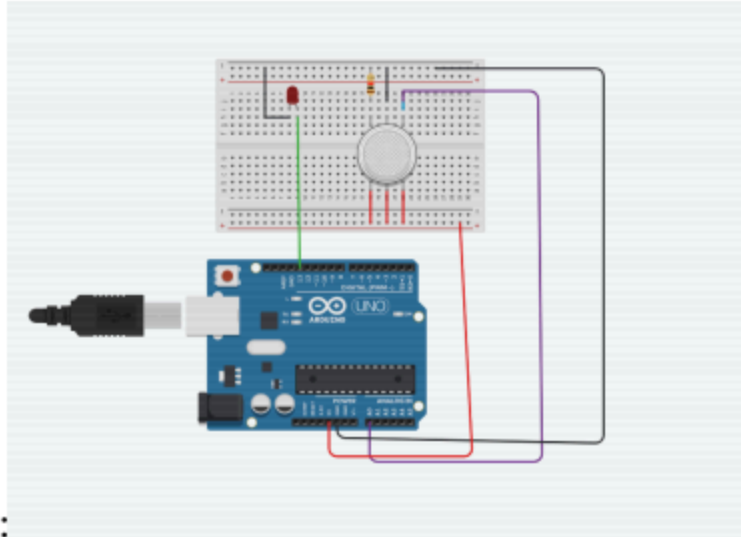
STEPS:

- Open Arduino IDE then goto tools and select appropriate Arduino board.
- Select necessary components and make circuit connections as shown in the diagram
- Make sure VCC and Ground pins connected properly to avoid any damage to Arduino board.
- write the code and run a simulator to get appropriate results.

Circuit Connections



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CODE

```
int LED = 13;

int MQ2pin = A0;

void setup() {
  Serial.begin(9600);
  pinMode(LED, OUTPUT); // Initialize the LED pin
}

void loop() {
  float sensorValue;
  sensorValue = analogRead(MQ2pin);

  if (sensorValue >= 250) {
    digitalWrite(LED, HIGH);
  }
}
```



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```

Serial.print(sensorValue);

Serial.println(" | GAS DETECTED");
} else {

digitalWrite(LED, LOW);

Serial.println("Sensor Value: ");

Serial.println(sensorValue);

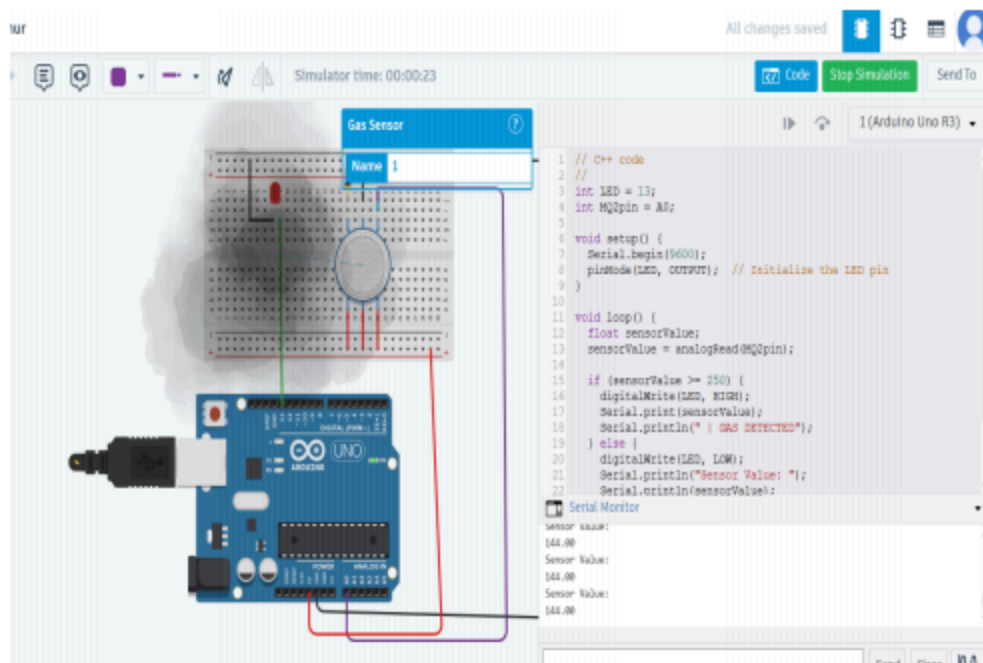
}

delay(1000); // Wait for 1 second
}

```

OUTPUT

The MQ2 gas sensor successfully detects gas concentration and triggers an alert (LED ON) when the threshold is exceeded.



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Connections Explained:

◆ *MQ2 Gas Sensor:*

It has 4 pins:

MQ2 Gas Sensor:

It has 4 pins:

Pin	Connected To
VCC	Arduino 5V (Red wire)
GND	Arduino GND (Black wire)
AOUT	Arduino A0 (Purple wire)
DOUT	(Not used in this setup)

- VCC and GND power the sensor.
- AOUT (Analog Output) reads varying gas concentrations and sends it to analog pin A0.

LED with Resistor:

- Anode (long leg) of LED is connected to **pin 13** of Arduino via **green wire**.
- Cathode (short leg) is connected to **GND through a resistor** (for current limiting, likely 220Ω).

This LED lights up when gas is detected based on the logic in your Arduino code.

CODE:

```
int LED = 13;

int MQ2pin = A0;

void setup() {
  Serial.begin(9600);

  pinMode(LED, OUTPUT); // Initialize the LED pin
```



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